

ECG Signal Classification Using 1D Convolutional Neural Network

This project focuses on classifying ECG heartbeat signals into Normal and Abnormal categories using a 1D Convolutional Neural Network (1D-CNN). The ECG signals were taken from the MIT-BIH Arrhythmia dataset, available on Kaggle in processed form.

ECG Waveform



1D CNN Architecture



The trained 1D-CNN model achieved approximately 92% accuracy in testing, demonstrating that deep learning can effectively distinguish between normal and abnormal heartbeat signals. This system can act as a supportive tool for early cardiovascular condition detection.